

DATA 308: Lab 1b

Working with data

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When working in a Quarto document I recommend:

- Start by rendering your document, so you know if there are any issues with the file.
- Render periodically as you work and inspect your rendered file, so you know your document and code are behaving as desired.

Getting your data

0. In the code chunk below load in the `tidyverse` package.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

1. In the code chunk below read in your data and save it into an object called `game_films`.

```
game_films <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday')
```

Rows: 439 Columns: 20

```
-- Column specification -----  
Delimiter: ","  
chr  (14): category, subcategory, title, director, release_date_raw, air_dat...  
dbl  (5): worldwide_box_office, rotten_tomatoes, metacritic, budget_low, bu...  
date (1): release_date
```

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

2. Use the `glimpse()` function to inspect your data. How many rows and columns are there? Note, I did not add a code chunk for you. You will first need to click the green “+C” button at the top right of this document to add a code chunk.

There are 20 columns and 439 rows.

```
glimpse(game_films)
```

Rows: 439

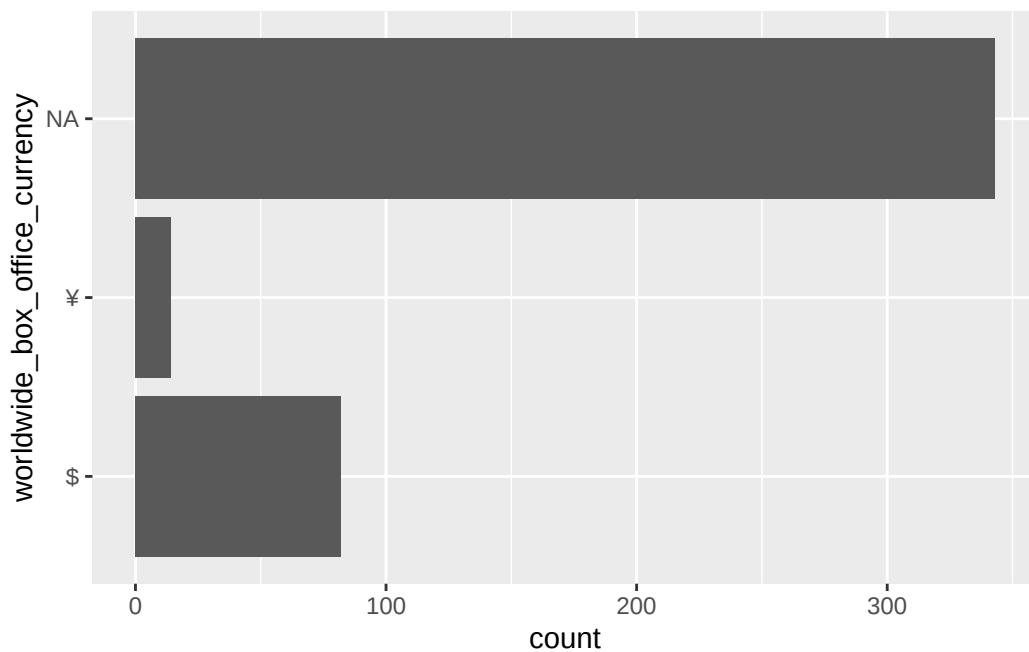
Columns: 20

```
$ category      <chr> "Theatrical releases", "Theatrical relea~  
$ subcategory   <chr> "English", "English", "English", "Englis~  
$ title         <chr> "Super Mario Bros.", "Double Dragon", "S~  
$ director      <chr> "Rocky Morton and Annabel Jankel", "Jame~  
$ release_date  <date> 1993-05-28, 1994-11-04, 1994-12-  
23, 199~  
$ release_date_raw <chr> "May 28, 1993", "November 4, 1994", "Dec~  
$ air_date_raw   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~  
$ worldwide_box_office_currency <chr> "$", "$", "$", "$", "$", "$", "$", "$", ~  
$ worldwide_box_office <dbl> 38912465, 4152699, 99423521, 124741822, ~  
$ rotten_tomatoes <dbl> 29, 12, 11, 47, 4, 10, 20, 36, 24, 3, 18~  
$ metacritic     <dbl> 35, 40, 34, 60, 11, 21, 33, 33, 43, 15, ~  
$ cinema_score   <chr> "B+", "F", "B-", "A-", "C+", "D", "B", "~  
$ distributor     <chr> "Buena Vista Pictures Distribution", "Gr~  
$ original_game_publisher <chr> "Nintendo", "Technōs Japan", "Capcom", "~  
$ budget_currency <chr> "$", "$", "$", "$", "$", "$", "$", "$", ~  
$ budget_low     <dbl> 4.20e+07, 7.80e+06, 3.50e+07, 2.00e+07, ~  
$ budget_high    <dbl> 4.80e+07, 7.80e+06, 3.50e+07, 2.00e+07, ~  
$ domestic_box_office <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~  
$ subject        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~  
$ network        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
```

Explore your data

3. Reproduce a plot from the lecture slides.

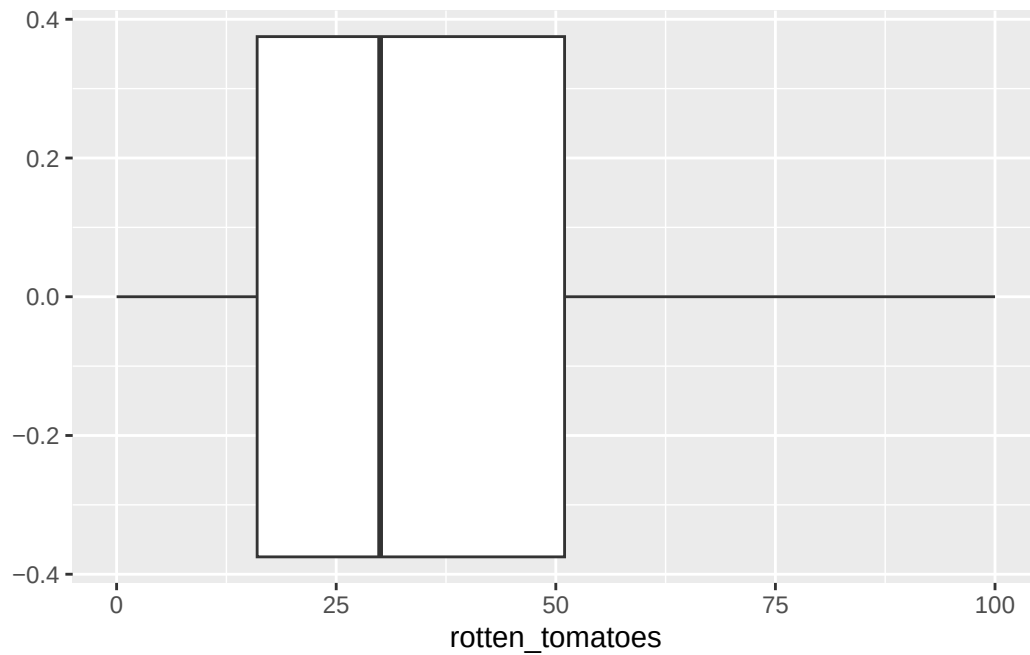
```
ggplot(game_films, aes(y = worldwide_box_office_currency)) +  
  geom_bar()
```



4. Create a box plot using `geom_boxplot()` for a numerical variable of your choice.

```
ggplot(game_films, aes(rotten_tomatoes)) +  
  geom_boxplot()
```

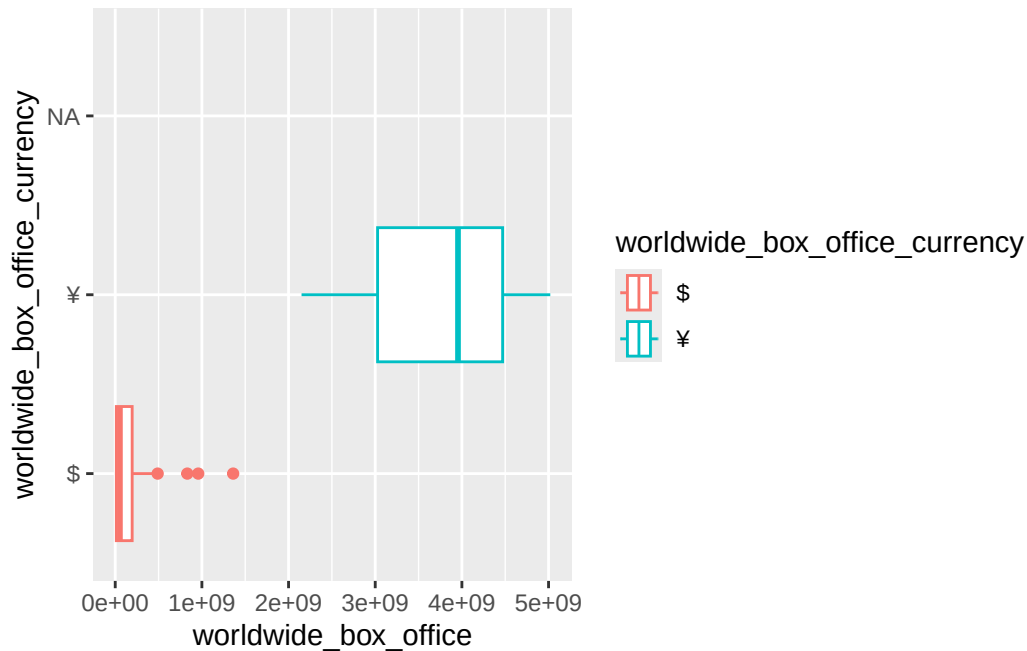
Warning: Removed 366 rows containing non-finite outside the scale range (``stat_boxplot()``).



5. Create a box plot of `worldwide_box_office` by `worldwide_box_office_currency`.
Hint: you can do this by specifying `worldwide_box_office_currency` for your y-axis.

```
ggplot(game_films, aes(x = worldwide_box_office, y = worldwide_box_office_currency, color = worldwide_box_office_currency)) +  
  geom_boxplot()
```

Warning: Removed 343 rows containing non-finite outside the scale range (``stat_boxplot()``).



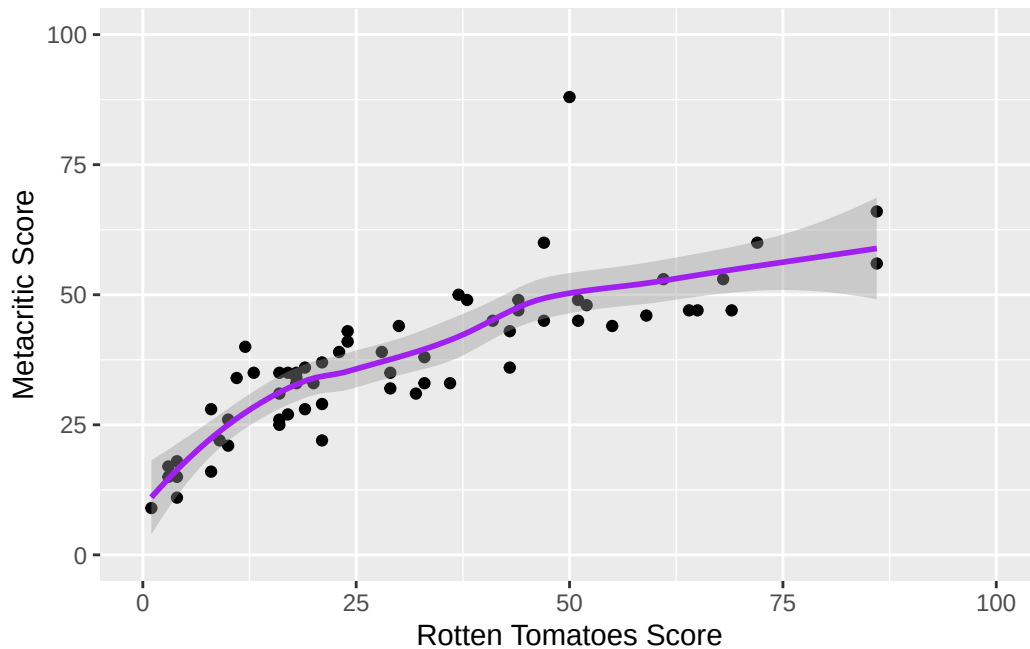
6. Create a scatterplot for two numerical variables of your choosing, and add a smoothed curve.

```
ggplot(game_films, aes(x = rotten_tomatoes, y = metacritic)) +
  geom_point() +
  geom_smooth(color = "purple") +
  coord_cartesian(xlim = c(0,100), ylim = c(0,100)) +
  labs(x = "Rotten Tomatoes Score",
       y = "Metacritic Score")
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 376 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 376 rows containing missing values or values outside the scale range
(`geom\_point()`).



7. Calculate the minimum, maximum, 25% quantile, and 75% quantile of `rotten_tomatoes` scores.

```
game_films |>
  summarise(
    min_rt = min(rotten_tomatoes, na.rm = TRUE),
    max_rt = max(rotten_tomatoes, na.rm = TRUE),
    q1_rt = quantile(rotten_tomatoes, 0.25, na.rm = TRUE),
    q3_rt = quantile(rotten_tomatoes, 0.75, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 4
  min_rt max_rt q1_rt q3_rt
  <dbl> <dbl> <dbl> <dbl>
1      0    100    16    51
```

Version control

I do recommend using GitHub for this class, but I will not require for homework assignments, just the final project. I recommend you practice by using a single GitHub repository for your labs and homework assignments for this class.

Here are [instructions I created for installing and configuring GitHub and RStudio](#). (Skip step 4)